

Q1 2026 Cross-Month EDA Summary

NYC TLC High-Volume For-Hire Vehicle (HVFHV) Trip Records

1. Q1 Dataset Overview

The Q1 2026 analysis combines **January, February and March 2026** HVFHV trip records to identify patterns that remain consistent across months and distinguish them from month-specific anomalies.

Month	Analytical Trips	Avg Daily Trips
January	20.94M	~675K
February	19.88M	~710K
March	22.06M	~712K
Q1 Total	62.87M	—

March recorded the highest monthly trip volume, while February had the lowest total volume because it contained only 28 days. Across the three months, the analysis identified several recurring patterns in **demand timing, provider structure, trip characteristics and trip economics**.

2. Recurring Demand Patterns

Evening Demand Peak

The strongest cross-month finding is the consistency of the **6 PM demand peak**.

Month	Peak Hour	Trips
January	6 PM	1,275,537
February	6 PM	1,196,467
March	6 PM	1,255,655

In all three months, demand was lowest around **3 AM** and increased through the day before reaching its highest level around the evening period.

This consistency makes **month + day + hour demand analysis** one of the strongest themes for the next stage of the project.

Demand Does Not Equal Trip Characteristics

A recurring pattern across the three months was that the **busiest hour was not necessarily the hour with the longest or farthest trips**.

Early-morning trips generally had longer average distances, while average trip duration tended to peak around the afternoon. In March, for example, 6 PM had the highest demand, but 3 PM had the highest average duration and 4 AM had the highest average distance.

This indicates that **trip volume, operational duration and trip value should be analysed as separate dimensions**.

3. Provider Market Structure

HV0003 consistently dominated the Q1 market.

Month	HV0003 Share	HV0005 Share
January	72.80%	27.20%

February	71.28%	28.72%
March	72.34%	27.66%
Q1	72.16%	27.84%

HV0003 therefore handled approximately **72% of all Q1 trips**.

The provider difference was not limited to trip volume. Across the monthly analyses, HV0003 also showed **higher average fare and generally higher average trip distance and driver pay**.

March provides a clear example:

- HV0003 average fare: **\$28.08**
- HV0005 average fare: **\$26.36**
- HV0003 average distance: **4.91 miles**
- HV0005 average distance: **4.51 miles**

HV0003's passenger-charge share in March was also **73.60%**, slightly above its 72.34% trip share.

The consistency across all three months suggests that **provider-level segmentation is a structural component of the Q1 market**, rather than a one-month observation.

4. Trip Characteristics & Economics

All three months showed a similar trip profile: **most trips were relatively short**, while a small number of very long trips created right-skewed distributions.

Median trip distance remained around **2.7–2.8 miles**, while median duration remained around the **mid-teen minutes**. This means median and percentile measures are more representative of a typical trip than averages alone.

February demonstrated a particularly clear relationship between distance and fare:

Trip Distance	Avg Base Fare
----------------------	----------------------

0–2 mi	\$12.90
2–5 mi	\$22.52
5–10 mi	\$35.72
10–15 mi	\$52.86
15–20 mi	\$76.16
20+ mi	\$116.74

The pattern shows that **longer trips are associated with substantially higher average fares**. February also showed that average tips increased as base fare increased, although this descriptive relationship does not establish causation.

5. Geographic & Segment-Level Patterns

Geographic concentration appeared consistently across the monthly analyses. Several pickup locations repeatedly appeared among the highest-volume areas.

February also highlighted a particularly strong destination concentration, with **Location 265 recording 821,845 drop-offs**. The repeated appearance of certain locations suggests that geographic demand should be examined at the **pickup, drop-off and route level** rather than only at the overall city level.

Airport trips also emerged as an important high-value segment. In January, airport-related trips represented **7.41% of trips**, but had much higher average fare, distance, duration and driver pay than non-airport trips.

Shared rides, on the other hand, remained a relatively small component of demand:

Month	Shared Request Rate
-------	---------------------

January	1.98%
February	1.94%
March	1.50%
Q1	1.80%

The declining monthly rate makes shared-ride behaviour worth examining as a potential **Q1 trend**, rather than treating it simply as a volume metric.

6. Data Quality & Analytical Considerations

The three months contained unusual observations including negative fares, extreme distances, timestamp inconsistencies and zero-distance/zero-time trips.

The consistent approach was to **investigate and flag unusual records rather than blindly delete them**. The core pickup-to-dropoff sequence remained valid across the analyses.

March additionally demonstrated the importance of accounting for **daylight-saving-time effects** when comparing recorded and timestamp-derived durations.

A further methodological consideration emerged from March's weekday analysis: raw weekday totals are not always directly comparable when weekdays occur different numbers of times within a month. Future SQL analysis should therefore use appropriate normalization where required.

7. Q1 Business Story & Analytical Direction

The three-month EDA produces a consistent overall picture:

Q1 HVFHV demand was highly time-dependent, geographically concentrated and dominated by HV0003. Most trips were short, while longer trips generated

substantially higher fare values. The busiest periods were not necessarily the periods with the longest or highest-value trips.

The strongest recurring themes are:

1. **Demand varies strongly by month, day and hour.**
2. **6 PM is consistently the major demand peak.**
3. **Early-morning trips tend to be longer despite lower demand.**
4. **HV0003 consistently handles roughly 70%+ of trips.**
5. **Provider differences extend beyond volume into trip characteristics and economics.**
6. **Most trips are short, with strongly right-skewed distance and duration distributions.**
7. **Geographic concentration is important at both pickup and drop-off locations.**
8. **Trip characteristics such as distance are closely associated with fare value.**
9. **Airport trips represent a smaller but substantially higher-value segment.**
10. **Shared rides remain a relatively small portion of overall demand.**
11. **Data-quality anomalies require investigation rather than automatic deletion.**

These findings provide the foundation for the **SQL layer**. SQL should now move beyond simply reproducing the Python EDA and instead answer **business questions that compare months, providers, trip economics, geography and operational patterns across Q1**.